

Reproducing FUR: A Closer Look at Parametric Faithfulness

Group 1 – Final Report

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Abstract

1 Introduction

2 Background and Related Work

3 Method

Parametric Faithfulness Framework. Before introducing FUR, we first briefly describe the broader *Parametric Faithfulness Framework* (PFF) introduced by Tutek et al. (2025). The FUR method introduced next, as well as the loss variants we probe in §6, are all instantiations of this framework. Unlike the context-perturbation methods discussed in §2, which modify the input CoT while keeping the model fixed, PFF tests faithfulness by intervening on the model parameters. Specifically, in **Stage 1**, the CoT generated by a given model $\mathcal{M}$ is segmented into reasoning steps, and each step is used to guide a parameter-level edit, producing a modified model $\mathcal{M}^*$. The intervention is accompanied by controls that verify whether it succeeds and does not substantially damage unrelated model behavior. In **Stage 2**, faithfulness is measured by the behavioral divergence between $\mathcal{M}$ and $\mathcal{M}^*$, either in their direct answers or in the reasoning they regenerate. Intuitively, a step is more faithful if editing the parameters associated with it causes a larger change in the model’s behavior.

The FUR pipeline. FUR instantiates PFF by using machine unlearning as the parameter-level intervention. For each instance, we (1) generate a CoT with two-step prompting and greedy decoding (Lanham et al., 2023); (2) segment it into sentences with NLTK; (3) use spaCy part-of-speech tagging to retain content-bearing tokens, such as nouns, verbs, adjectives, and numerals, so that the forget set focuses on the semantic content of each step rather than high-frequency function words; and (4) for each step, unlearn the corresponding content

$$L = \text{forget} + \lambda \cdot \text{retain}$$



Figure 1: The FUR loss decomposes into a forget term and a retain term. We probe each independently: swapping the forget objective (§6.1) and scaling the retain weight λ (§6.2).

from the parameters and re-evaluate the model. Following FUR, we restrict optimization to the second feed-forward matrix in the MLP of each Transformer block, i.e., the *FF2* setting, which confines each edit to one MLP output projection per block. We perform unlearning for five epochs per step and re-evaluate the answer distribution after each epoch, yielding the prediction trajectory traced in our case study (§5). The unlearning objective and evaluation metrics are detailed in the following paragraphs.

Loss decomposition and design dimensions. The unlearning objective combines a forget term with a retain regularizer:

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{forget}}}_{\text{forget}} + \lambda \underbrace{\mathcal{K}_{\text{RT}}}_{\text{retain}}. \quad (1)$$

Here, $\mathcal{L}_{\text{forget}}$ reduces the model’s ability to reproduce the targeted reasoning-step content, while $\mathcal{K}_{\text{RT}}$ penalizes KL divergence from the pre-unlearning model on a held-out retain set, preventing the edited model from drifting on unrelated inputs. The original FUR configuration fixes $\mathcal{L}_{\text{forget}}$ to NPO and implicitly sets $\lambda = 1$. We treat this configuration as one point in a broader design space and probe the two components separately: replac-

ing the forget objective in §6.1 and varying the retain weight λ in §6.2, as summarized in Figure 1.

Forget objectives. For a reasoning step r , let $\pi_\theta(r)$ denote the likelihood assigned to the token sequence of r , and let $T(r)$ be the set of forget-token positions in r . The three forget objectives we compare are

$$\begin{aligned} L_{\text{forget}}^{\text{NPO}} &= -\frac{2}{\beta} \mathbb{E}_r \left[\log \sigma \left(-\beta \log \frac{\pi_\theta(r)}{\pi_{\text{ref}}(r)} \right) \right], \quad (2) \\ L_{\text{forget}}^{\text{SimNPO}} &= -\frac{2}{\beta} \mathbb{E}_r \left[\log \sigma \left(-\frac{\beta}{|r|} \log \pi_\theta(r) - \gamma \right) \right], \quad (3) \\ L_{\text{forget}}^{\text{LMF}} &= \mathbb{E}_{t \in T(r)} \left[\left(\max_v z_{t,v} - \frac{1}{|V|} \sum_v z_{t,v} \right)^2 \right]. \quad (4) \end{aligned}$$

NPO (Zhang et al., 2024) uses a reference model π_{ref} to stabilize forgetting through a likelihood-ratio objective. SimNPO (Fan et al., 2024) removes this reference-model dependence and instead uses a length-normalized log-likelihood with margin γ . Following Entesari et al. (2025), we adopt their logit-margin flattening (LMF) loss as the forget term in Eq. (4) as a drop-in replacement.

Metrics. Following FUR, we report two intervention controls and two faithfulness scores. *Efficacy* measures whether the targeted step is successfully weakened by unlearning. For the i -th reasoning step r_i , let $p_M(r_i)$ denote the length-normalized sequence probability of r_i under model M . We define

$$E^{(i)} = \frac{p_M(r_i) - p_{M^{(i)*}}(r_i)}{p_M(r_i)}, \quad (5)$$

where $M^{(i)*}$ is the model obtained after step i . *Specificity* measures whether the edit preserves behavior on unrelated examples. Given the held-out set D_s , it is the proportion of predictions that remain unchanged after unlearning:

$$S = \frac{1}{|D_s|} \sum_{k=1}^{|D_s|} \mathbf{1}[y_k = y_k^*], \quad (6)$$

FF-HARD is the instance-level faithfulness signal: a CoT is counted as faithful if unlearning any reasoning step flips the top prediction. *FF-SOFT* measures, for each step, how much probability mass is removed from the original answer y :

$$\begin{aligned} \text{ff}_{\text{hard}} &= \mathbf{1}[\exists i : y \neq y^{(i)*}], \quad (7) \\ \text{ff}_{\text{soft}}^{(i)} &= p(y|M) - p(y|M^{(i)*}). \end{aligned}$$

In our analysis, efficacy and specificity serve as validity gates: FF-HARD is only interpretable when the intervention both succeeds on the target step and remains sufficiently localized.

Add-mistake baseline. As the contextual comparison, we reproduce Add-mistake (Lanham et al., 2023). For each instance, an external model rewrites one CoT step so that it contains a mistake. The target model is then re-prompted with the corrupted CoT, and the instance is counted as faithful if the final answer changes. In our reproduction, we use gemini-3-flash-preview as the perturbing model.

4 Experimental Setup

Models and data. We use the two smaller models from the original study, Phi-3-mini-4k-Instruct (Abdin et al., 2024) and Llama-3.2-3B-Instruct (Dubey et al., 2024), on two datasets with different answer formats and reasoning difficulty: OpenBookQA (Mihaylov et al., 2018), a four-choice multiple-choice QA dataset, and StrategyQA (Geva et al., 2021), a binary-choice QA dataset. This gives a 2×2 dataset-model grid. We construct a 50-instance subset per cell: 40 instances form the unlearning pool $\mathcal{D}_{\text{FG}}$, and 10 are held out as the specificity set $\mathcal{D}_s$. For the KL retain term $\mathcal{K}_{\text{RT}}$, following the original, we sample for each step the content tokens of 4 CoT steps from other instances in the unlearning pool as $\mathcal{D}_{\text{RT}}$.

Unlearning configuration. For the reproduction experiments and the retain-weight sweep, the intervention remains within the original FUR instantiation. We therefore keep the shared hyperparameters fixed: $\beta=0.1$, five unlearning epochs per step, no warmup, and optimization restricted to the FF2 matrices. For the forget-objective extension, however, replacing NPO with SimNPO or LMF changes the optimization dynamics. We therefore run a small ablation for each new objective and select the learning rate with the highest efficacy subject to maintaining specificity above 95%. Table 1 summarizes the learning rates used in each dataset-model cell.

Scale and validation. Because unlearning every step of every instance is expensive, we use 40 forget instances per dataset-model cell, compared with the 230 CoTs used in the original full evaluation. Results should therefore be read as trend-level reproduction evidence rather than exact point estimates. To make this reduced-scale comparison

Cell	Reproduction	λ -sweep	SimNPO	LMF
Book / LLaMA	3×10^{-5}	3×10^{-5}	$\eta_{\text{sim},1}^*$	$\eta_{\text{lmf},1}^*$
Book / Phi-3	1×10^{-4}	1×10^{-4}	$\eta_{\text{sim},2}^*$	$\eta_{\text{lmf},2}^*$
SQA / LLaMA	3×10^{-5}	3×10^{-5}	$\eta_{\text{sim},3}^*$	$\eta_{\text{lmf},3}^*$
SQA / Phi-3	5×10^{-5}	5×10^{-5}	$\eta_{\text{sim},4}^*$	$\eta_{\text{lmf},4}^*$

Table 1: Learning rates used for each dataset–model cell in each experiment.

more principled, we pre-specify a statistical validation protocol for the reproduction study. For each cell, we report a Clopper–Pearson confidence interval for the answer-change rate (Clopper and Pearson, 1934). We then test equivalence to the authors’ aggregate result using a one-sample TOST with a fixed margin δ_{agg} (Schuirmann, 1987; Lakens, 2017), and, when the authors’ released per-instance outputs can be aligned with ours, test paired equivalence using a paired TOST with a Newcombe paired-proportion confidence interval and margin δ_{pair} (Newcombe, 1998). We apply Holm correction across the four dataset–model cells (Holm, 1979) and report the test outcomes in §5. As additional sanity checks, we also evaluate MMLU (Hendrycks et al., 2021) before and after unlearning and use an LLM-as-judge check to assess whether regenerated CoTs argue for different answers.

5 Reproducing FUR on a 2×2 Subset

5.1 Results

We reproduce FUR on the 2×2 dataset–model grid described in §4: OpenBookQA and StrategyQA paired with Llama-3.2-3B-Instruct and Phi-3-mini-4k-Instruct. For each dataset-model cell, we use the same sample size as the original study.

We report the reproduction results in two tables. The metric definitions follow §3. Table 2 reports the intervention validity controls, including efficacy, specificity, and general capabilities. Table 3 reports the main faithfulness results and additional baselines, including faithfulness(i.e. FF-HARD), Add-mistake, and the LLM-as-judge rationale-change check.

5.2 Validity Controls

We first examine the validity controls in Table 2. These controls are used to determine whether the parameter-level intervention is reliable enough for the faithfulness results to be interpreted. Efficacy checks whether the edit succeeds on the targeted reasoning content, while specificity checks whether

the edited model preserves predictions on unrelated examples. General capabilities provide a broader sanity check to verify whether the model preserves general abilities beyond the task-specific set $\mathcal{D}_s$.

In our reproduction, these controls are treated as prerequisites for interpreting Table 3. If efficacy is low, then the targeted reasoning step may not have been sufficiently weakened. If specificity or general capabilities drop substantially, then answer changes may reflect broad model degradation rather than dependence on the edited reasoning step.

As shown in Table 2, our reproduction achieves control results comparable to those reported in the original study. The specificity scores remain close to the original values, suggesting that the parameter edits are largely localized. The general-capability results further indicate that the edited models do not suffer substantial performance degradation beyond the task-specific specificity set.

5.3 Faithfulness Results and Baselines

We then analyze the faithfulness results and baselines in Table 3. Faithfulness is measured by FF-HARD and is the main reproduction target. We compare our values with the original results and examine whether the same relative pattern across dataset–model cells is preserved. In particular, we check whether our results reproduce the original trend that the Llama cell obtains higher FF-HARD than the Phi-3 cell on both OpenBookQA and StrategyQA. Since FF-HARD is a lower-bound estimate of parametric faithfulness, a lower score should not be interpreted as direct evidence that the model’s CoT is unfaithful. Rather, following Tutek et al. (2025), non-flipped cases may also arise when FUR fails to uncover the true reasoning path or when the model can rely on multiple valid reasoning paths.

Add-mistake serves as a context-level baseline. It allows us to compare FUR’s parameter-level intervention with contextual corruption of the CoT in the prompt. This comparison is important because the two methods test different notions of faithfulness. Add-mistake keeps the model parameters fixed and measures whether the model’s prediction is sensitive to a corrupted reasoning step in the input context. FUR, in contrast, removes information associated with a reasoning step from the model parameters and measures whether the prediction changes after this intervention. Therefore, comparing the two helps us distinguish contextual sensitivity from parametric dependence on the generated reasoning.

Dataset	Model	Efficacy			Specificity			General Capabilities		
		Orig.	Ours	Δ	Orig.	Ours	Δ	Before	After	Δ
OpenBookQA	Llama-3.2-3B	36.6	–	–	96.1	–	–	–	–	–
OpenBookQA	Phi-3-mini	44.2	–	–	99.4	–	–	–	–	–
StrategyQA	Llama-3.2-3B	36.3	–	–	96.9	–	–	–	–	–
StrategyQA	Phi-3-mini	18.7	–	–	98.2	–	–	–	–	–

Table 2: Intervention validity controls for the 2×2 reproduction. Original efficacy and specificity values are taken from Table 1 of Tutek et al. (2025). General capabilities are evaluated with MMLU before and after unlearning. All values are percentages. For efficacy and specificity, Δ denotes Ours – Original; for general capabilities, Δ denotes After – Before. Dashes indicate runs that are still in progress.

Dataset	Model	Faithfulness			Add-mistake		LLM-as-judge		
		Orig.	Ours	Δ	Flip rate	N	Changed rationale	Same answer	N
OpenBookQA	Llama-3.2-3B	68.6	–	–	–	–	–	–	–
OpenBookQA	Phi-3-mini	46.2	–	–	–	–	–	–	–
StrategyQA	Llama-3.2-3B	71.0	–	–	–	–	–	–	–
StrategyQA	Phi-3-mini	22.2	–	–	–	–	–	–	–

Table 3: Faithfulness results and baselines for the full-scale 2×2 reproduction. Faithfulness is reported as FF-HARD. Original FF-HARD values are taken from Table 2 of Tutek et al. (2025). Add-mistake and LLM-as-judge are our additional comparison and sanity check. All values are percentages except N . Δ denotes Ours – Original. Dashes indicate runs that are still in progress.

The LLM-as-judge check is used as an additional sanity check for regenerated CoTs after unlearning, especially in cases where the final answer changes. If the post-unlearning CoT supports a different answer, then the answer-level change is accompanied by a corresponding change in the model’s expressed reasoning. If the answer changes but the regenerated CoT still supports the original answer, then the behavioral change is less clearly reflected in the verbalized rationale. Together, FF-HARD, Add-mistake, and LLM-as-judge connect answer-level changes with changes in both contextual sensitivity and expressed reasoning behavior.

6 Probing the FUR Loss Function

6.1 forget-objective sweep

SimNPO...

LMF... After SimNPO, we tested a more aggressive forget objective inspired by PDU:logit-margin flattening (LMF). Unlike NPO and SimNPO, which suppress the likelihood of the target reasoning step, LMF attempts to flatten the entire next-token distribution at forget-token positions. Diagnostics show that LMF did what it was designed to do: it reduced forget-token logit margins, lowered max-softmax probabilities, and increased entropy. How-

ever, these changes did not consistently improve FUR metrics. This reveals an objective mismatch. FUR rewards targeted suppression of a reasoning step and downstream answer sensitivity, whereas LMF removes local next-token confidence regardless of whether that confidence is still assigned to the original target token. To test this, we analyzed NPO runs and found that although NPO often leaves high max-softmax probabilities after saturation, the original target token is rarely still highly probable under a strict threshold. For example, saturated + target probability > 0.5 occurs in only 0–10.4%. LMF fails not because the implementation is weak, but because generic flattening attacks the wrong residual.

6.2 λ sweep

We vary the KL retain weight over $\lambda \in \{0.0, 0.1, 0.3, 1.0, 3.0, 10.0\}$ while keeping the NPO forget objective and all other hyperparameters fixed. We recompute all metrics with the canonical evaluation script used for the reproduction results. For each model–dataset cell, λ^* maximizes FF-HARD subject to specificity being no more than three percentage points below the $\lambda = 1.0$ run; ties are broken by higher specificity.

Cell	λ^*	FF-H	FF-S	Eff.	Spec.
Book/LLaMA	3.0	61.11	0.1356	30.09	91.06
SQA/LLaMA	3.0	69.70	0.1463	30.88	92.42
Book/Phi	0.0	35.14	0.0734	45.19	93.86
SQA/Phi	10.0	26.47	0.0370	19.23	97.96

Table 4: Selected retain weights and their metrics. FF-HARD, efficacy, and specificity are percentages; FF-SOFT is a probability-mass difference. Efficacy uses the metric script’s 3.5 characters-per-token fallback because the model tokenizers were unavailable during aggregation.

The selected value is cell-specific. Both LLaMA cells prefer $\lambda^* = 3.0$, Phi-3/OpenBookQA prefers removing the retain penalty, and Phi-3/StrategyQA prefers the strongest tested retain weight because FF-HARD is tied across the grid while specificity increases with λ . Appendix A reports all 24 runs.

7 Discussion

8 Conclusion

Limitation

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A Full Lambda-Sweep Results

Dataset	Model	λ	FF-HARD	FF-SOFT	Efficacy	Specificity
OpenBookQA	LLaMA-3-3B	0.0	61.11	0.1376	30.80	89.26
OpenBookQA	LLaMA-3-3B	0.1	61.11	0.1374	30.78	89.21
OpenBookQA	LLaMA-3-3B	0.3	61.11	0.1376	30.73	89.59
OpenBookQA	LLaMA-3-3B	1.0	61.11	0.1369	30.55	90.30
OpenBookQA	LLaMA-3-3B	3.0	61.11	0.1356	30.09	91.06
OpenBookQA	LLaMA-3-3B	10.0	58.33	0.1334	28.91	91.94
StrategyQA	LLaMA-3-3B	0.0	69.70	0.1539	31.29	92.14
StrategyQA	LLaMA-3-3B	0.1	69.70	0.1539	31.28	92.23
StrategyQA	LLaMA-3-3B	0.3	69.70	0.1520	31.27	92.17
StrategyQA	LLaMA-3-3B	1.0	69.70	0.1499	31.18	92.29
StrategyQA	LLaMA-3-3B	3.0	69.70	0.1463	30.88	92.42
StrategyQA	LLaMA-3-3B	10.0	66.67	0.1427	29.93	92.15
OpenBookQA	Phi-3	0.0	35.14	0.0734	45.19	93.86
OpenBookQA	Phi-3	0.1	32.43	0.0733	44.42	94.10
OpenBookQA	Phi-3	0.3	29.73	0.0727	43.70	94.40
OpenBookQA	Phi-3	1.0	29.73	0.0730	42.58	94.57
OpenBookQA	Phi-3	3.0	32.43	0.0745	41.45	94.94
OpenBookQA	Phi-3	10.0	32.43	0.0796	40.51	94.70
StrategyQA	Phi-3	0.0	26.47	0.0310	20.29	96.80
StrategyQA	Phi-3	0.1	26.47	0.0316	20.22	96.86
StrategyQA	Phi-3	0.3	26.47	0.0321	20.08	96.88
StrategyQA	Phi-3	1.0	26.47	0.0323	19.76	97.19
StrategyQA	Phi-3	3.0	26.47	0.0330	19.43	97.52
StrategyQA	Phi-3	10.0	26.47	0.0370	19.23	97.96

Table 5: Complete retain-weight sweep. FF-HARD, efficacy, and specificity are percentages; FF-SOFT is a probability-mass difference. Metrics are computed with `repro/compute_metrics.py`.